

★ PRIMER · UNIT 10

Before the Lecture: How to Fool Yourself

Unit 10 · Honest Backtesting

Watch this first if the Sharpe ratio and cross-validation are new

Twenty minutes of groundwork — how a strategy is scored, why shuffling time is cheating, and how many tries it takes before luck looks like skill.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Sharpe and CV are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 10 Sits

The course builds one pipeline; here is this unit's stone

Unit 9

done

Structure found without labels.

Unit 10

you are here

Testing that does not lie to you.

Unit 11

next

Portfolios that do not break.

This is the unit that decides whether anything in the previous nine was real.

What a Backtest Actually Is

A backtest is a practice exam you wrote, sat, and marked yourself — on questions you had already seen.

What it can show

- The rule was computable.
- The logic runs end to end.
- No obvious bug.

What it cannot show

- That it will work ahead.
- That the edge was real.
- That you did not peek.

Every technique in this unit exists to narrow the gap between those two columns.

The Words the Lecture Assumes

Five terms, and both halves of the unit open up

Word	What it means	Here
Sharpe ratio	return per unit of risk	the standard score
Annualise	scale to a year	multiply by root 252
Leakage	future information sneaking in	the silent killer
Cross-validation	train on some rows, test on the rest	standard, and broken here
Overlap	two labels sharing days	why purging is needed

Leakage is the one word to remember. Most impressive backtests are leaking somewhere.

Three Splitters, One Honesty Ladder

Each rung removes a way the previous one lied

Splitter	What it does	Still wrong because
K-fold	shuffles rows freely	time gets scrambled
Walk-forward	respects time order	labels still overlap
Purged k-fold	drops overlapping rows	nothing — this is the standard

Combinatorial purged CV goes one further, and gives a distribution instead of one number.

Why Shuffling Time Is Cheating

Shuffle the rows and a model can train on Thursday to predict Wednesday. It will do very well.

In most fields

- Rows are independent.
- Order carries nothing.
- Shuffling is harmless.

In finance

- Rows are a sequence.
- Order is the information.
- Shuffling leaks the future.

This single mistake explains more spectacular backtests than every other cause combined.

A Sharpe Ratio by Hand

Four monthly returns — compute the mean, the spread, and the ratio

Month	Return %	Deviation	Squared
Jan	+2.0	+1.0	1.00
Feb	−1.0	−2.0	4.00
Mar	+3.0	+2.0	4.00
Apr	+0.0	−1.0	1.00
mean +1.0	—	—	sum 10.00

Variance = $10/4 = 2.50$, so $sd = 1.58$. Monthly Sharpe = $1.0 / 1.58 = 0.63$.

Annualising That Number

One multiplication, and one very common abuse of it

The scaling

- Monthly Sharpe 0.63.
- Multiply by root 12.
- Annual Sharpe 2.19.

The catch

- Root-time assumes independence.
- Returns are not independent.
- The number flatters you.

An annualised Sharpe from four months of data is arithmetic, not evidence

What Pure Luck Can Manufacture

Try enough strategies and one will look brilliant by chance alone

One strategy

- Sharpe 2.0 is impressive.
- Hard to get by luck.
- Worth a second look.

Two thousand tried

- The best will hit 2.0 anyway.
- Luck, not skill.
- You will never know which.

The Deflated Sharpe Ratio asks how good a number must be, given how many you tried.

Purging a Timeline

Three moves, and the lecture works one timeline in full

1 · Find

the overlap

A label started Monday and ended Friday touches the whole week.

2 · Purge

the training side

Drop training rows whose labels overlap the test window.

3 · Embargo

just after

Drop a few more rows right after the test, for serial correlation.

You throw data away on purpose. That discomfort is the price of an honest number.

The Leakage Rogues' Gallery

Four ways the future gets into the past, all of them quiet

Leak	How it happens	Symptom
Shuffled time	standard k-fold on a series	impossibly high accuracy
Scaling first	mean and sd from all data	small, steady inflation
Overlapping labels	one event in train and test	optimistic, hard to spot
Restated data	today's values, back-dated	perfect on old periods

Every one of these makes the result better. None of them ever makes it worse.

Notation You Will Meet

Four symbols carry the whole unit

Symbol	Read it as	It stands for
SR	Sharpe ratio	mean return divided by its standard deviation
$\sqrt{252}$	root 252	the annualising factor for daily data
DSR	deflated Sharpe	the Sharpe after paying for how many you tried
PBO	P B O	the probability the backtest was overfit

252 is trading days in a year. Use 12 for monthly, 52 for weekly.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>r.mean() / r.std()</code>	the Sharpe ratio you just did by hand
<code>* np.sqrt(252)</code>	annualise it, with the caveat above
<code>TimeSeriesSplit(n_splits=5)</code>	respect time order — the first honest rung
<code>Pipeline([('sc', StandardScaler()), ...])</code>	scale inside the fold, not before it
<code>cross_val_score(pipe, X, y, cv=tscv)</code>	score without leaking

Putting the scaler in a Pipeline is not tidiness — it is what stops the second leak.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 11

1

"A Sharpe of 3 proves the strategy works"

It proves it fitted the past. Ask how many strategies were tried before this one.

2

"I scaled the features first, which is good practice"

Scaling before the split uses test-set statistics. Scale inside each fold instead.

3

"Cross-validation is cross-validation"

Standard k-fold shuffles time. On financial data it is not validation at all.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1	Mean 1.0, sd 1.58. What is the Sharpe?	0.63
2	You scale the whole dataset, then split. What leaked?	test statistics
3	Why drop training rows near the test window?	overlapping labels

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It shows what luck can manufacture

How good a Sharpe ratio pure chance produces once you have tried enough strategies.

2

It rebuilds cross-validation for time

Purging, embargo, and combinatorial purged CV — the honesty ladder, rung by rung.

3

It ends in code

The problem set scores a strategy, purges a timeline by hand, and computes a deflated Sharpe.

Next: Unit 10 lecture — How Not to Fool Yourself